

that can be used to “download” the code and print any new drug in that comes in the market. For our junkies, no, you will not have an endless supply of “Your Favourite medicine”(if you know what we mean) as there will be some restrictions as to what can and cannot be printed.

Print your food

Imagine your mom ‘printing’ chicken and then putting it in the oven to cook! The technology hasn’t reached that far, but is surprisingly close. Don’t be surprised if you see bio-printed labels on the meat that you buy in the supermarket. A company called Modern Meadow is gearing up to do just that, starting with a slightly less controversial animal product: leather. It’s not as frightening as it might seem, and it’s not really instant “Star-Trek” style.

Bio-printing uses 3D printing technology, inspired by the common inkjet printer, refined to handle delicate cells and other laboratory substances. Since the recent development of ways to produce vascular structures (blood vessels) in bio-printed tissue, a simple, juicy steak containing types of muscle cells, fat and related structures won’t be that hard to create. But the question here is that will it be tasty enough to eat? If yes, then consumers may start seeing meat produced this way as a true meat-substitute.

The process to create it is straightforward. Cells are cultured and grown in quantity in the laboratory. They’re combined in a tissue pattern and herded together into a three dimensional structure by a bio-printer. Special techniques are used to provide structure and support while the cells link up naturally into the final product.

How would you order a steak using this process? A special software provides a vision of the final product, from the fat and



Print your food and eat it

marbling to the type of meat required. The bioprinter combines the materials under computer’s direction. The process of individual cells joining up into meat still takes a few weeks, but after that the job is done, and the meat can be prepared and turned over to the chef. Bon appétit!

3D Printing in space

NASA has come up with a plan to print buildings on the moon. Obviously, there are some technical issues to address, the first being, the size of the presently available printers needs to be scaled so as



Mars Rovers

to be able to print buildings. Research is underway as you read this sentence. CAD/CAM, an automated method of design and manufacturing, is already used to produce most consumer goods, such as clothing, electronics, and some cars.

Professor Khoshnevis aims to apply similar methodology to the archaic method of constructing houses and he’s already built and tested a system that can print a hollow concrete wall that can withstand 10,000 psi, compared to the 3,000 of a traditionally built wall of the same size that takes more materials and time. Additionally, more complex structures and molding could be printed for less. That is, your house could have domes, arches and hexagonal honeycomb rooms for the same cost as a plain square box house and every house on the block could be custom made.

Back on Mars, Desert RATS (Research and Technology Studies) a NASA division is working on a prototype made up of 70 basic parts (all additive manufactured) and the size of a Hummer that can support two humans in a pressurised cabin. The rovers must be strong and light, durable production-grade thermoplastics that can withstand the harsh environment of other worlds. 

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